

Alpha Diversity

Muluh

Community diversity

Group-wise comparisons

- **Index:** observed
- **Group variable:** meal_group
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

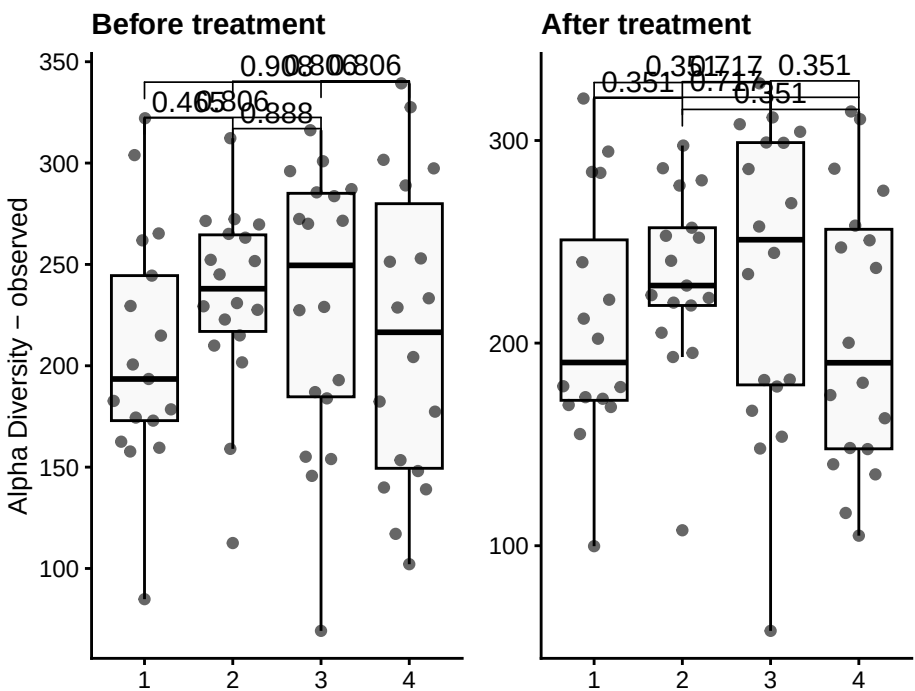


Table of the top findings

Across-group

Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

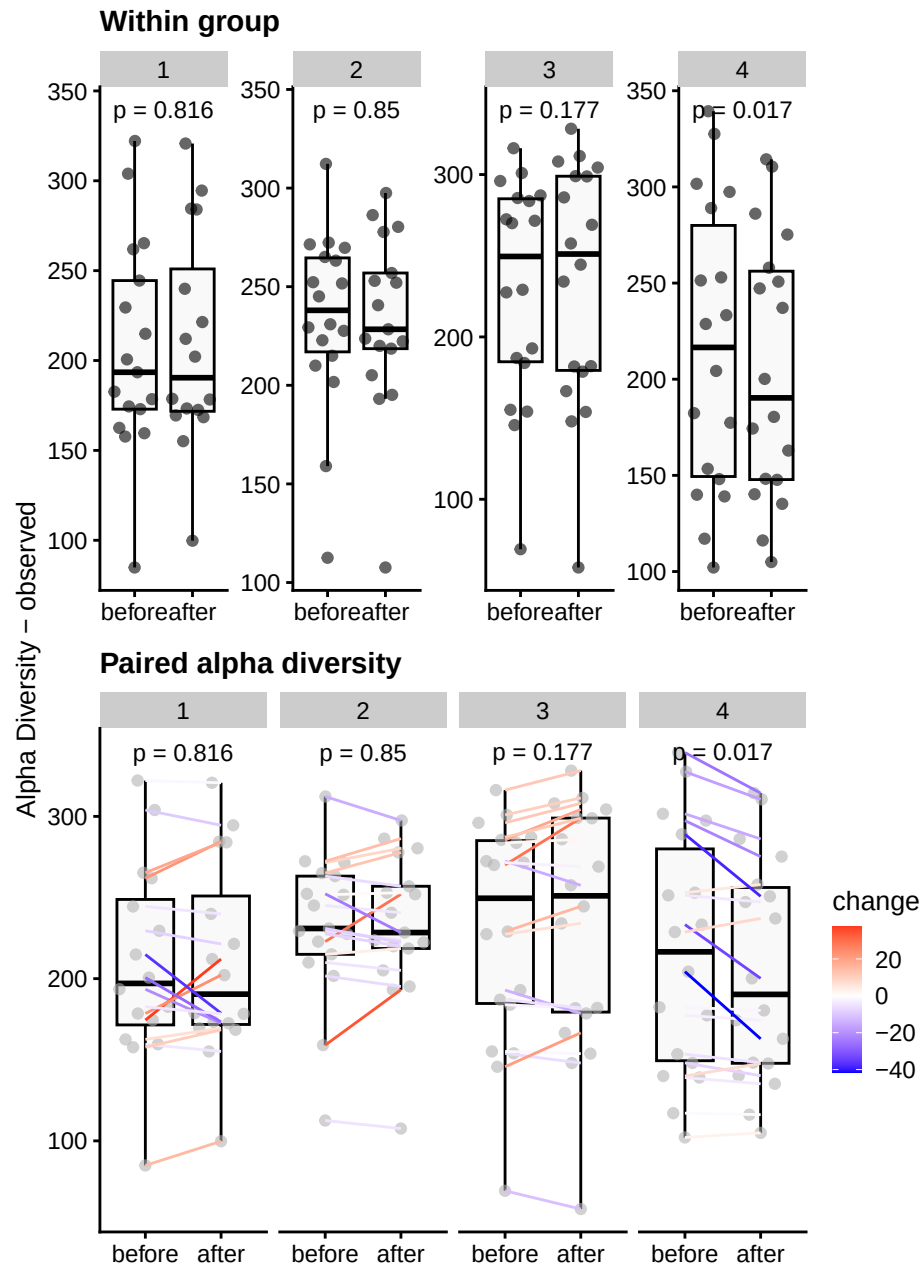
group1	group2	mean1	mean2	log2FC	p.adj
1	2	206.44	234.00	0.18	0.465
1	3	206.44	229.33	0.15	0.806
1	4	206.44	215.83	0.06	0.908
2	3	234.00	229.33	-0.03	0.888
2	4	234.00	215.83	-0.12	0.806
3	4	229.33	215.83	-0.09	0.806

After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	209.69	232.85	0.15	0.351
1	3	209.69	233.87	0.16	0.351
1	4	209.69	204.97	-0.03	0.717
2	3	232.85	233.87	0.01	0.717
2	4	232.85	204.97	-0.18	0.351
3	4	233.87	204.97	-0.19	0.351



Within-group

Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Characteristic	before			after		
	Beta	95% CI ^I	p-value	Beta	95% CI ^I	p-value
(Intercept)	151	67, 236	<0.001	144	59, 229	0.001
meal_group			0.4			0.3
1	—	—		—	—	
2	32	-11, 75	0.14	28	-16, 72	0.2
3	28	-15, 71	0.2	30	-14, 73	0.2
4	10	-32, 53	0.6	-3.9	-47, 39	0.9
age	1.0	-0.44, 2.5	0.2	1.2	-0.25, 2.7	0.10

^ICI = Confidence Interval

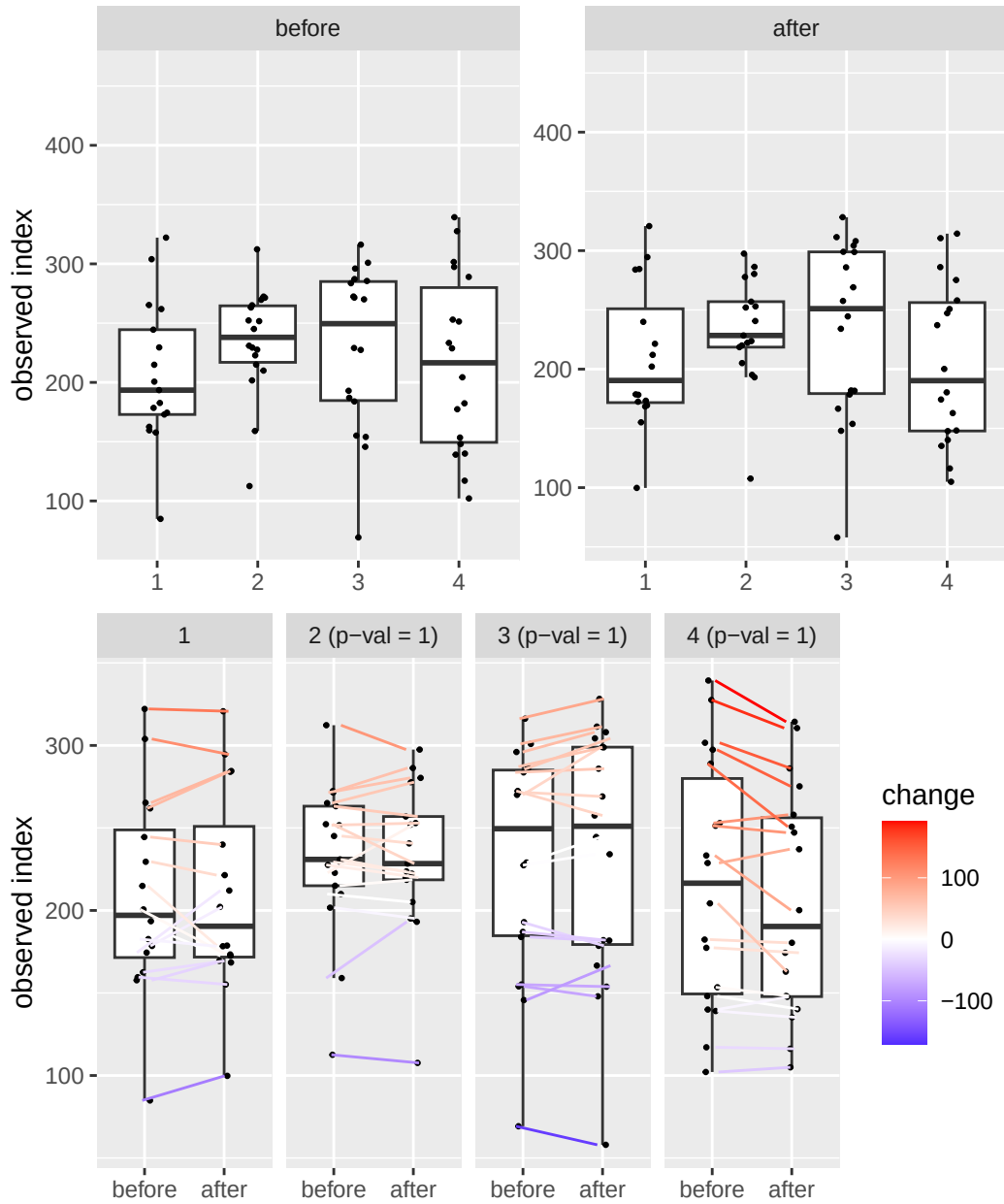
Paired test of before and after	Beta	95% CI ^I	p-value
(Intercept)	147	-Inf, Inf	<0.001
meal_group			0.4
1	—	—	
2	28	-Inf, Inf	>0.9
3	28	-Inf, Inf	>0.9
4	2.1	-Inf, Inf	>0.9
age	1.2	0.14, 2.2	0.025

^ICI = Confidence Interval

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

meal_group	mean1	mean2	log2FC	p.value
1	206.44	209.69	0.06	0.816
2	234.00	232.85	0.07	0.850
3	229.33	233.87	0.02	0.177
4	215.83	204.97	-0.06	0.017

Alternate analyses



Data Summary

class: TreeSummarizedExperiment
dim: 1700 140

```

metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```

Version info

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R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

```

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Matrix products: default
BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

```

```

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

```

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time zone: Europe/Helsinki
tzcode source: system (glibc)

```

```

attached base packages:
[1] stats4      grid        stats      graphics  grDevices  utils      datasets
[8] methods    base

```

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other attached packages:
[1] kableExtra_1.4.0      vegan_2.6-8
[3] lattice_0.22-6        permute_0.9-7

```

[5] lubridate_1.9.3	forcats_1.0.0
[7] stringr_1.5.1	purrr_1.0.2
[9] readr_2.1.5	tibble_3.2.1
[11] tidyverse_2.0.0	tidyr_1.3.1
[13] sechm_1.13.1	ComplexHeatmap_2.21.1
[15] scatter_1.33.4	scuttle_1.15.4
[17] quarto_1.4.4	patchwork_1.3.0
[19] parameters_0.24.0	multtest_2.61.0
[21] miaTime_0.1.21	miaViz_1.15.4
[23] ggraph_2.2.1	mia_1.15.6
[25] TreeSummarizedExperiment_2.13.0	Biostrings_2.73.2
[27] XVector_0.45.0	SingleCellExperiment_1.27.2
[29] MultiAssayExperiment_1.31.5	SummarizedExperiment_1.35.3
[31] Biobase_2.65.1	GenomicRanges_1.57.1
[33] GenomeInfoDb_1.41.2	IRanges_2.39.2
[35] S4Vectors_0.43.2	BiocGenerics_0.51.3
[37] MatrixGenerics_1.17.0	matrixStats_1.4.1
[39] gridExtra_2.3	gtsummary_2.0.4
[41] ggsignif_0.6.4	ggpubr_0.6.0
[43] ggplot2_3.5.1	DT_0.33
[45] dplyr_1.1.4	cardx_0.2.2
[47] ANCOMBC_2.8.0	

loaded via a namespace (and not attached):

[1] randomcoloR_1.1.0.1	gld_2.6.6
[3] nnet_7.3-19	TH.data_1.1-2
[5] vctrs_0.6.5	energy_1.7-12
[7] digest_0.6.37	png_0.1-8
[9] shape_1.4.6.1	proxy_0.4-27
[11] Exact_3.3	registry_0.5-1
[13] rbiom_1.0.3	ggrepel_0.9.6
[15] bayestestR_0.15.0	parallelly_1.38.0
[17] mediation_4.5.0	MASS_7.3-61
[19] reshape2_1.4.4	foreach_1.5.2
[21] withr_3.0.2	xfun_0.49
[23] ggfun_0.1.6	survival_3.7-0
[25] doRNG_1.8.6	commonmark_1.9.1
[27] memoise_2.0.1	ggbeeswarm_0.7.2
[29] emmeans_1.10.4	systemfonts_1.1.0
[31] gmp_0.7-5	tidytree_0.4.6
[33] zoo_1.8-12	GlobalOptions_0.1.2
[35] gtools_3.9.5	V8_5.0.1
[37] Formula_1.2-5	datawizard_0.13.0

[39]	httr_1.4.7	rstatix_0.7.2
[41]	globals_0.16.3	ps_1.8.0
[43]	rstudioapi_0.17.1	UCSC.utils_1.1.0
[45]	generics_0.1.3	base64enc_0.1-3
[47]	processx_3.8.4	curl_5.2.3
[49]	zlibbioc_1.51.1	ScaledMatrix_1.13.0
[51]	polyclip_1.10-7	ca_0.71.1
[53]	GenomeInfoDbData_1.2.13	SparseArray_1.5.41
[55]	xtable_1.8-4	doParallel_1.0.17
[57]	evaluate_1.0.1	S4Arrays_1.5.10
[59]	hms_1.1.3	irlba_2.3.5.1
[61]	labelled_2.13.0	colorspace_2.1-1
[63]	readxl_1.4.3	magrittr_2.0.3
[65]	later_1.3.2	viridis_0.6.5
[67]	ggtree_3.13.1	DECIPHER_3.1.4
[69]	cowplot_1.1.3	class_7.3-22
[71]	Hmisc_5.2-0	pillar_1.9.0
[73]	nlme_3.1-166	iterators_1.0.14
[75]	decontam_1.25.0	compiler_4.4.2
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[81]	tokenizers_0.3.0	minqa_1.2.8
[83]	plyr_1.8.9	crayon_1.5.3
[85]	abind_1.4-8	gridGraphics_0.5-1
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[89]	bit_4.5.0	rootSolve_1.8.2.4
[91]	sandwich_3.1-1	codetools_0.2-20
[93]	multcomp_1.4-26	BiocSingular_1.21.4
[95]	slam_0.1-54	e1071_1.7-16
[97]	lmom_3.2	GetoptLong_1.0.5
[99]	tidytext_0.4.2	splines_4.4.2
[101]	markdown_1.13	circlize_0.4.16
[103]	Rcpp_1.0.13-1	sparseMatrixStats_1.17.2
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[107]	utf8_1.2.4	clue_0.3-65
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[119]	Matrix_1.7-0	svglite_2.1.3
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[133]	rmarkdown_2.29	scales_1.3.0
[135]	cards_0.4.0	broom_1.0.7
[137]	coda_0.19-4.1	insight_1.0.0
[139]	carData_3.0-5	rpart_4.1.23
[141]	farver_2.1.2	tidygraph_1.3.1
[143]	mgcv_1.9-1	yaml_2.3.10
[145]	foreign_0.8-87	cli_3.6.3
[147]	lifecycle_1.0.4	mvtnorm_1.3-2
[149]	bluster_1.15.1	backports_1.5.0
[151]	BiocParallel_1.39.0	timechange_0.3.0
[153]	gtable_0.3.6	rjson_0.2.23
[155]	parallel_4.4.2	ape_5.8
[157]	SnowballC_0.7.1	CVXR_1.0-15
[159]	jsonlite_1.8.9	seriation_1.5.6
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[163]	yulab.utils_0.1.8	BiocNeighbors_1.99.1
[165]	RcppParallel_5.1.9	janeaustrinr_1.0.0
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[169]	tinytex_0.54	glue_1.8.0
[171]	broom.helpers_1.17.0	treeio_1.29.1
[173]	boot_1.3-31	igraph_2.1.1
[175]	R6_2.5.1	broom.mixed_0.2.9.6
[177]	labeling_0.4.3	Rmpfr_1.0-0
[179]	cluster_2.1.8	rngtools_1.5.2
[181]	aplot_0.2.3	nloptr_2.1.1
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[185]	tidyselect_1.2.1	vipor_0.4.7
[187]	htmlTable_2.4.3	xml2_1.3.6
[189]	ggforce_0.4.2	car_3.1-3
[191]	future_1.34.0	rsvd_1.0.5
[193]	munsell_0.5.1	furrr_0.3.1
[195]	data.table_1.16.2	htmlwidgets_1.6.4
[197]	RColorBrewer_1.1-3	rlang_1.1.4
[199]	lmerTest_3.1-3	ggnewscale_0.5.0
[201]	fansi_1.0.6	beeswarm_0.4.0